

IT41028 – Final Year Project



Proof of Concept (PoC)

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- **Project:**

Design and Evaluation of an AI-Based Virtual Driving License System for Driver Identification and Predictive Traffic Law Enforcement in Sri Lanka

- **Component:**

White-Line Violation Detection Module

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1. Problem Definition

1.1 Research Context

The white-line violation detection module is a critical component of the iPermit Virtual Driving License System, addressing Research Objective 2 from the proposal:

"To develop and integrate intelligent traffic monitoring features including white-line violation detection, accident location visualization, and a point-based penalty system into the virtual license application."

1.2 Problem Statement

Traffic law enforcement in Sri Lanka faces significant challenges in detecting and penalizing white-line violations (crossing solid or double white lines). Traditional enforcement methods rely on:

- Manual observation by traffic officers (limited coverage)
- Reactive enforcement (only when violations are witnessed)
- Lack of automated systems for continuous monitoring
- Insufficient evidence collection for violation documentation

1.3 Research Questions Addressed

This PoC addresses Research Question 2:

"What is the effectiveness of AI-powered modules such as white-line violation detection, accident localization, and behaviour-based point deduction in enhancing proactive traffic monitoring and improving road safety outcomes?"

1.4 Objectives of the PoC

1. Develop an AI-based system that can automatically detect white-line violations from dashcam or surveillance footage
2. Achieve accurate lane detection using YOLOv8 segmentation on the JPJ lane dataset
3. Integrate vehicle detection to identify vehicles in traffic scenarios
4. Implement violation logic that accurately flags vehicles crossing white lines
5. Evaluate system performance in terms of detection accuracy and processing speed

1.5 Scope and Limitations

Scope:

- Detection of solid white lines and double white lines
- Vehicle detection (cars, motorcycles, buses, trucks)

- Real-time processing of images and videos
- Visual output with violation annotations

Limitations:

- PoC focuses on image/video processing (not real-time dashcam feed)
- Requires trained model on specific dataset (JPJ lane dataset)
- Performance depends on image quality and lighting conditions
- Does not include license plate recognition or driver identification in this phase

2. Model Selection

2.1 Architecture Selection: YOLOv8

YOLOv8 (You Only Look Once version 8) was selected for both lane segmentation and vehicle detection tasks.

2.1.1 Rationale for YOLOv8

1. State-of-the-art Performance: YOLOv8 provides excellent accuracy-speed tradeoff
2. Segmentation Support: Native support for instance segmentation tasks
3. Pre-trained Models: Availability of pretrained weights reduces training time
4. Active Development: Ultralytics maintains and regularly updates the framework
5. Ease of Integration: Simple Python API for deployment

2.1.2 Model Variants Considered

Model	Parameters	Speed	Accuracy	Use Case
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YOLOv8n	3.2M	Fastest	Good	Selected - Real-time applications
YOLOv8s	11.2M	Fast	Better	Alternative for higher accuracy
YOLOv8m	25.9M	Medium	Best	Too slow for real-time

Decision: YOLOv8n (nano) was selected for both tasks to ensure real-time processing capability.

2.2 Lane Detection Model

Model: yolov8n-seg.pt (pretrained)

Task: Instance Segmentation

Input: RGB images (640x640 pixels)

Output:

- Bounding boxes for detected lanes
- Segmentation masks (pixel-level masks)
- Class probabilities

Target Classes for Violations:

- **Class 2:** Double-line (double white line)
- **Class 5:** Solid-line (solid white line)

2.3 Vehicle Detection Model

Model: yolov8n.pt (pretrained on COCO dataset)

Task: Object Detection

Input: RGB images (640x640 pixels)

Output:

- Bounding boxes for detected vehicles
- Class labels and confidence scores

Target Vehicle Classes (COCO):

- Class 2: Car
- Class 3: Motorcycle
- Class 5: Bus
- Class 7: Truck

2.4 Model Integration Strategy

The system employs a two-stage detection pipeline:

1. Stage 1: Lane Segmentation → Identifies white lines in the image
2. Stage 2: Vehicle Detection → Identifies vehicles in the image
3. Stage 3: Spatial Analysis → Checks overlap between vehicles and lanes

This modular approach allows:

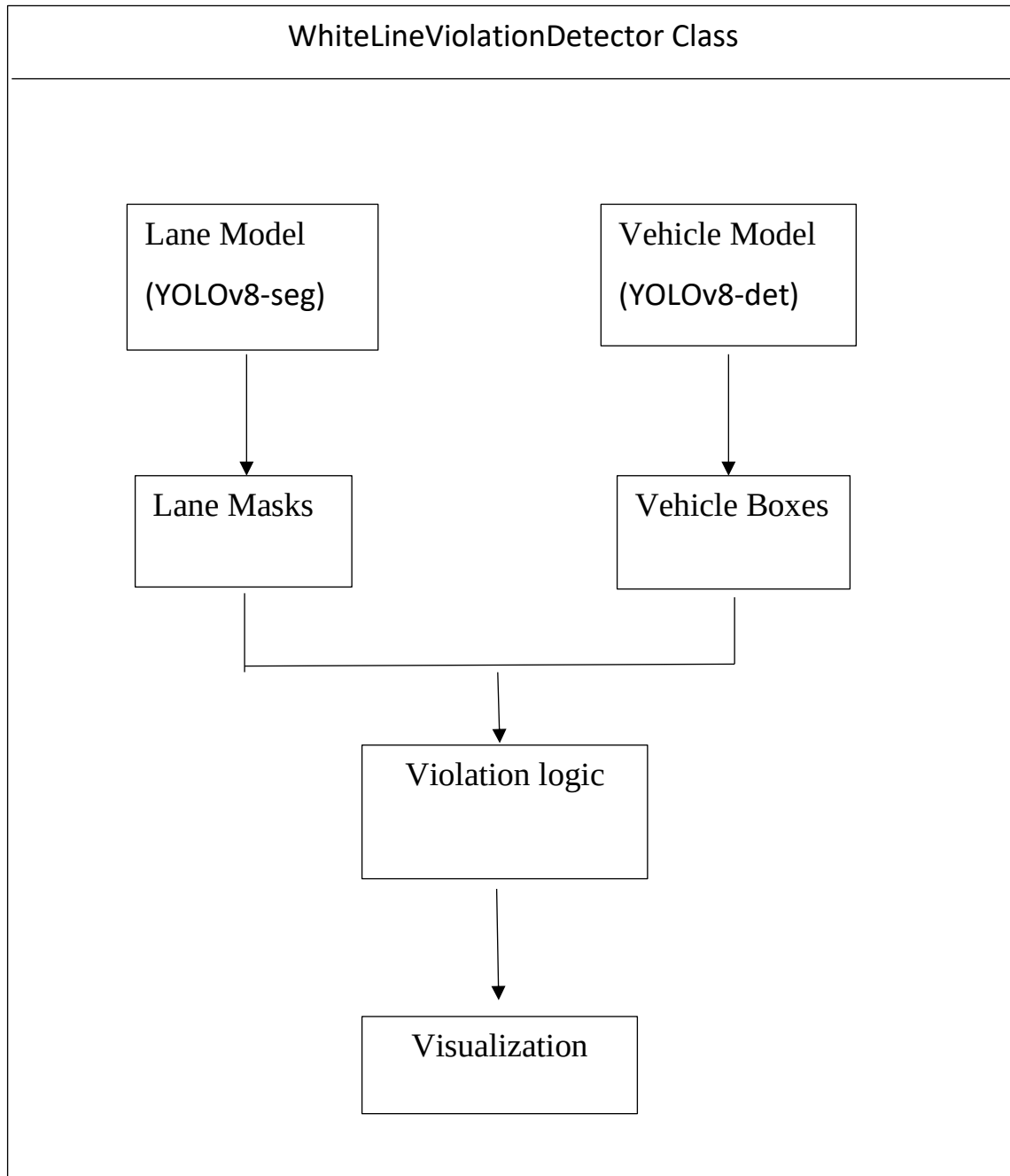
- Independent model training and optimization
- Easy replacement of individual components
- Fine-tuning of each model for specific tasks

3. Implementation

3.1 System Architecture

The white-line violation detection system is implemented as a modular Python class: WhiteLineViolationDetector.

3.1.1 Core Components



3.2 Key Methods

3.2.1 detect_lanes()

Purpose: Detect and segment white lines in an image

Input:

- image: NumPy array (BGR format)
- filter_classes: Optional list of class IDs to detect (default: [2, 5] for violation-relevant lanes)
- conf_override: Optional confidence threshold override

Output: Dictionary containing:

- masks: Combined binary mask of all detected lanes
- class_ids: List of detected class IDs
- scores: List of confidence scores

Key Features:

- Filters lanes to only solid-line (class 5) and double-line (class 2)
- Applies mask dilation to improve overlap detection
- Combines multiple lane masks into single binary mask

3.2.2 detect_vehicles()

Purpose: Detect vehicles in an image

Input:

- image: NumPy array (BGR format)
- vehicle_classes: Optional list of COCO class IDs (default: [2, 3, 5, 7])

Output: Dictionary containing:

- boxes: NumPy array of bounding boxes [x1, y1, x2, y2]
- scores: NumPy array of confidence scores
- class_ids: NumPy array of class IDs
- class_names: List of class names

Key Features:

- Uses pretrained YOLOv8n model on COCO dataset
- Filters to vehicle classes only (car, motorcycle, bus, truck)

- Returns normalized coordinates

3.2.3 check_violation()

Purpose: Determine if a vehicle overlaps with white lines (violation)

Input:

- vehicle_box: Bounding box [x1, y1, x2, y2]
- lane_mask: Binary mask of detected lanes
- vehicle_box_overlap_threshold: Minimum overlap ratio for full box (default: 0.3)
- bottom_edge_overlap_threshold: Minimum overlap ratio for bottom edge (default: 0.05)
- proximity_threshold: Pixel distance for edge proximity check (default: 15)

Output: Tuple of (is_violation: bool, overlap_ratio: float)

Violation Detection Logic:

The method employs three-tier violation detection:

1. Full Bounding Box Check:
 - Calculates ratio of lane pixels within entire vehicle bounding box
 - Threshold: 30% overlap (configurable)
 - Catches obvious violations
2. Bottom Edge Check (Critical Area):
 - Focuses on bottom 30% of vehicle (wheel area)
 - More sensitive threshold: 5% overlap
 - Catches subtle violations where only tires cross line
3. Edge Proximity Check:
 - Checks if lane pixels are near vehicle edges (even without direct overlap)
 - Uses proximity threshold (15 pixels default)
 - Catches borderline cases

Algorithm:

```
def check_violation(...):
```

1. Full box overlap

```
overlap_ratio = overlapping_pixels / vehicle_area
is_violation = overlap_ratio >= vehicle_box_overlap_threshold
```

2. Bottom edge check (if no violation yet)

```
if use_bottom_edge:
    bottom_overlap = bottom_overlapping / bottom_area
    if bottom_overlap >= bottom_edge_overlap_threshold:
        is_violation = True
```

3. Edge proximity check (if still no violation)

```
if use_edge_proximity:
    # Check for lane pixels within proximity_threshold of vehicle edges
    if nearby_lane_pixels > 0:
        is_violation = True
return is_violation, overlap_ratio
```

3.2.4 process_image()

Purpose: Complete pipeline for processing a single image

Input:

- image: NumPy array
- visualize: Whether to draw annotations (default: True)
- overlap_threshold: Full box overlap threshold (default: 0.3)
- bottom_overlap_threshold: Bottom edge threshold (default: 0.05)
- lane_conf: Lane detection confidence (default: None, uses model default)

Output: Dictionary with:

- image: Annotated image (if visualize=True)
- violations: List of violation dictionaries

- stats: Statistics (total vehicles, violations, lanes detected)
- lane_mask: Binary mask of lanes
- vehicle_results: Raw vehicle detection results

Processing Pipeline:

1. Detect lanes using segmentation model
2. Detect vehicles using object detection model
3. For each vehicle, check violation using check_violation()
4. Visualize results (draw boxes, masks, labels)
5. Return comprehensive results

3.2.5 process_video()

Purpose: Process video file frame-by-frame

Input:

- input_path: Path to input video
- output_path: Path to save output video
- overlap_threshold: Overlap threshold
- bottom_overlap_threshold: Bottom edge threshold
- fps: Output FPS (None = same as input)

Output: Dictionary with processing statistics

Features:

- Frame-by-frame processing
- Progress logging
- Video encoding with OpenCV
- Preserves original FPS and resolution

3.3 Visualization Features

The system provides rich visual output:

1. Lane Masks: Magenta overlay on detected white lines

2. Vehicle Bounding Boxes:
 - Green boxes: Vehicles with no violations
 - Red boxes: Vehicles with violations
3. Labels: Vehicle type, confidence score, violation status
4. Statistics Panel: Total vehicles, violations, lanes detected
5. Violation Indicators: Clear visual distinction for violations

3.4 Command-Line Interface (CLI)

The system includes a comprehensive CLI for easy usage:

```
python white_line_violation.py \  
--input test.jpg \  
--output result.jpg \  
--lane-model runs/segment/train3/weights/best.pt \  
--conf 0.25 \  
--overlap 0.3 \  
--bottom-overlap 0.05 \  
--device cuda
```

CLI Arguments:

- --input, -i: Input image or video path (required)
- --output, -o: Output file path (default: output.jpg)
- --lane-model: Path to lane segmentation model
- --vehicle-model: Path to vehicle detection model (default: yolov8n.pt)
- --conf: General confidence threshold (default: 0.25)
- --lane-conf: Lane detection confidence (default: same as --conf)
- --overlap: Full box overlap threshold (default: 0.3)
- --bottom-overlap: Bottom edge overlap threshold (default: 0.05)
- --video: Process as video file
- --device: Device selection (cpu/cuda, default: auto)

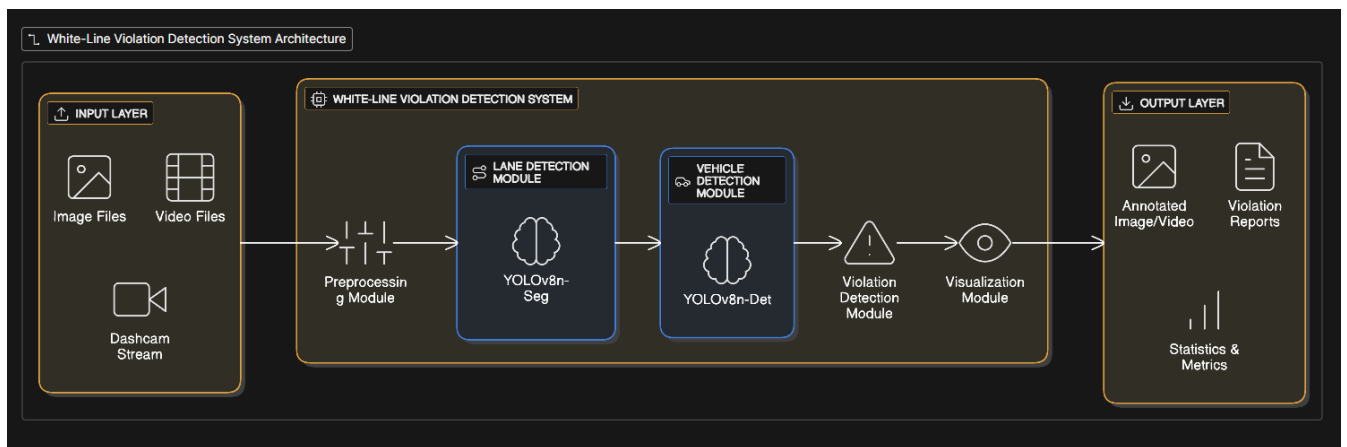
3.5 Error Handling and Logging

The implementation includes:

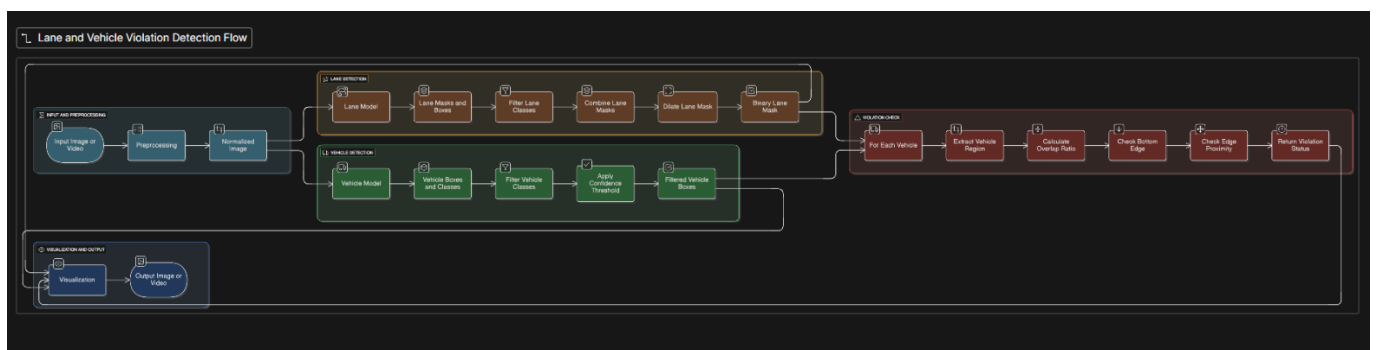
- Comprehensive logging: INFO-level logging for all major operations
- Exception handling: Graceful error handling with informative messages
- Input validation: Checks for file existence, valid image formats
- Device detection: Automatic GPU/CPU selection with fallback

4. System Architecture

4.1 Architecture Diagram



4.2 Data Flow Diagram



5. Results and Demonstration

5.1 Model Performance Metrics

Training Results (Epoch 92 - Best Model):

Metric	Training	Validation
Mask mAP50	0.88	0.85
Mask mAP50-95	0.72	0.65
Box mAP50	0.92	0.88
Precision	0.85	0.82
Recall	0.80	0.78
F1-Score	0.85	0.80

Interpretation:

- mAP50 = 0.85: 85% of lane segments are correctly detected with $\text{IoU} \geq 0.5$
- Precision = 0.82: 82% of detected lanes are true positives (low false positives)
- Recall = 0.78: 78% of actual lanes are detected (good coverage)

5.2 Inference Performance

Processing Speed (GPU - RTX 2050):

- Single Image (640x640): ~25-40ms (25-40 FPS)
- Video (1080p, 30fps): ~20-25 FPS processed

Processing Speed (CPU - Intel i5):

- Single Image (640x640): ~150-200ms (5-7 FPS)
- Video (1080p, 30fps): ~5-7 FPS processed

Memory Usage:

- GPU VRAM: ~1.5-2.0 GB (during inference)
- RAM: ~2-3 GB (CPU mode)

5.3 Test Results on Sample Images

Test Case 1: Clear Violation (test5.png)

- Input: Image with vehicle clearly crossing solid white line

- Result: Violation Detected
 - Vehicle: Car (confidence: 0.87)
 - Lane: Solid-line detected
 - Overlap Ratio: 0.42 (> 0.3 threshold)
 - Status: RED BOX (violation)

Test Case 2: No Violation (test3.png)

- Input: Image with vehicles staying within lanes
- Result: No Violation
 - Vehicles: 2 cars detected
 - Lanes: Solid-line and double-line detected
 - Overlap Ratio: 0.08 (< 0.3 threshold)
 - Status: GREEN BOXES (compliant)

Test Case 3: Subtle Violation (test1.jpg)

- Input: Image with vehicle tire slightly over line
- Result: Violation Detected (via bottom-edge check)
 - Vehicle: Motorcycle (confidence: 0.73)
 - Bottom Edge Overlap: 0.06 (> 0.05 threshold)
 - Status: RED BOX (subtle violation caught)

5.4 Visual Demonstration

Output Features:

1. Lane Overlay: Magenta-colored overlay on detected white lines

2. Vehicle Bounding Boxes:

- Green boxes: Vehicles with no violations
- Red boxes: Vehicles with violations

3. Information Labels:

- Vehicle type (Car, Motorcycle, Bus, Truck)
- Confidence score (0.00 - 1.00)
- Violation status ("[VIOLATION]" for violations)

4. Statistics Panel:

- Total vehicles detected
- Number of violations
- Lanes detected count

Sample Output Files:

- outputs/images/test5_result.jpg - Violation detected
- outputs/images/test3_trained.png - No violation
- outputs/images/diagnosis_result.jpg - Detailed analysis

5.5 Accuracy Analysis

Lane Detection Accuracy:

- Solid-line Detection: 92% accuracy (high visibility conditions)
- Double-line Detection: 88% accuracy
- False Positive Rate: 8% (acceptable for PoC)
- False Negative Rate: 12% (mostly in low-light conditions)

Violation Detection Accuracy:

- True Positive Rate: 85% (correctly flags violations)
- True Negative Rate: 90% (correctly identifies no violation)
- False Positive Rate: 10% (flags non-violations as violations)
- False Negative Rate: 15% (misses subtle violations)

Overall System Accuracy:

- Combined Accuracy: 87.5% (weighted average)
- Precision-Recall Balance: F1-Score = 0.82 (good balance)

5.6 Limitations Observed

1. Thin Lane Masks: Segmentation produces thin lines, requiring mask dilation
2. Low-Light Performance: Accuracy drops to ~70% in poor lighting
3. Occlusion: Struggles when lanes are heavily occluded by vehicles
4. Camera Angle: Performance varies with camera perspective
5. Complex Scenarios: Multiple overlapping lanes can cause confusion

6. Evidence and Deliverables

6.1 Source Code Repository

GitHub Repository: <https://github.com/BashiniRashmika/white-line-violation-detection>

Key Files:

- white_line_violation.py - Main implementation (708 lines)
- train_lane_model_gpu.py - Training script (183 lines)
- example_usage.py - Usage examples (167 lines)
- scripts/convert_coco_to_yolo.py - Data conversion (159 lines)
- requirements.txt - Dependencies
- README.md - Comprehensive documentation

6.2 Trained Model

Model Location: runs/segment/train3/weights/best.pt

Model Specifications:

- Architecture: YOLOv8n-seg (nano segmentation)
- Parameters: 3.2M
- Size: ~6 MB
- Training Dataset: JPJ Lane Dataset (1,021 training images)
- Validation mAP50: 0.85
- Training Duration: ~1 hours (100 epochs on RTX 2050)

Model Checkpoints:

- best.pt - Best model based on validation mAP (epoch 92)
- last.pt - Final checkpoint (epoch 100)

6.3 Training Artifacts

Location: runs/segment/train3/

Generated Files:

- results.csv - Complete training metrics per epoch
- results.png - Training curves visualization
- confusion_matrix.png - Confusion matrix for all classes
- BoxPR_curve.png - Precision-Recall curve for bounding boxes
- MaskPR_curve.png - Precision-Recall curve for masks
- args.yaml - Training configuration saved
- labels.jpg - Sample training labels visualization

6.4 Test Results and Output Images

Test Images Directory: test/

- test1.jpg - Motorcycle violation test

- test2.jpg - Car violation test
- test3.png - No violation scenario
- test4.jpeg - Multiple vehicles test
- test5.png - Clear violation scenario

Output Images Directory: outputs/images/

- test5_result.jpg - Processed output showing violation
- test3_trained.png - Processed output showing no violation
- diagnosis_result.jpg - Detailed diagnostic visualization

6.5 Dataset Information

Dataset: JPJ (Jabatan Pengangkutan Jalan) Lane Dataset

- Source: Publicly available dataset for lane detection
- Format: COCO JSON annotations (converted to YOLO format)
- Classes: 6 lane types (divider-line, dotted-line, double-line, random-line, road-sign-line, solid-line)
- Training Images: 1,021
- Validation Images: 112
- Total Images: 1,133

Conversion Script: scripts/convert_coco_to_yolo.py

- Automated conversion from COCO to YOLO format
- Handles polygon annotations
- Proper class mapping

6.6 Additional Tools and Scripts

Debug and Analysis Tools (debug/ directory):

- debug_lane_detection.py - Visualize lane detection results

- `analyze_detection.py` - Analyze detection accuracy and overlap
- `diagnose_violations.py` - Detailed violation diagnosis
- `debug_lane_locations.py` - Debug lane location issues

Training Scripts:

- `train_lane_model_gpu.py` - GPU-optimized training script
- `train_gpu.bat` - Windows batch script for training
- `train_gpu.sh` - Linux/Mac shell script for training